

6 February 2023

# A Differential Delay Cell for Ring Oscillators

Puneet Singh<sup>1</sup>, Mayank Kumar SINGH, Mahendra Sakare

1. Indian Institute of Technology Ropar

## Abstract

This paper presents a differential delay cell (DDC) for a 3-stage current-starved ring voltage-controlled oscillator (CSRVCO). In the proposed delay cell, pass transistors and a pair of back-to-back transistors are connected in a symmetric structure that helps get less jittery output by providing similar charging and discharging paths from the load capacitor. The current flows from similar paths with the same type and number of transistors in the proposed DDC-based CSRVCO. For verifying the DDC operation, the 3-stage CSRVCO using the proposed DDC is implemented with a standard 65nm CMOS technology with 1V supply. Post-layout simulation results confirm the correct operation of the proposed DDC using the CSRVCO with minimal jitter at the output. The proposed CSRVCO can be tuned from 0.88 GHz to 2.76 GHz. The phase noise of -94.54 dBc/Hz was measured at 1MHz offset frequency for the center frequency of 2.76 GHz. The CSRVCO was designed using the proposed DDC occupied area 1198.43  $\mu\text{m}^2$ , and dissipated power is 5.19mW.

## Keywords

communication, networking and broadcast technologies, current starved ring vco, delay, delay cell, inverter, phase noise, ring vco, vco

# A differential delay cell for ring oscillators

Puneet Singh, Mayank Kumar Singh, Mahendra Sakare

*Dept. of Electrical Engineering*  
*Indian Institute of Technology Ropar*  
Rupnagar, Punjab, India  
Email: puneet.19eez0003@iitrpr.ac.in

**Abstract**—This paper presents a differential delay cell (DDC) for a 3-stage current-starved ring voltage-controlled oscillator (CSRVCO). In the proposed delay cell, pass transistors and a pair of back-to-back transistors are connected in a symmetric structure that helps get less jittery output by providing similar charging and discharging paths from the load capacitor. The current flows from similar paths with the same type and number of transistors in the proposed DDC-based CSRVCO. For verifying the DDC operation, the 3-stage CSRVCO using the proposed DDC is implemented with a standard 65 nm CMOS technology with 1 V supply. Post-layout simulation results confirm the correct operation of the proposed DDC using the CSRVCO with minimal jitter at the output. The proposed CSRVCO can be tuned from 0.88 GHz to 2.76 GHz. The phase noise of  $-94.54 \frac{dBc}{Hz}$  was measured at 1 MHz offset frequency for the center frequency of 2.76 GHz. The CSRVCO was designed using the proposed DDC occupied area  $1198.43 \mu m^2$ , and dissipated power is 5.19 mW.

**Index Terms**—VCO, Inverter, Ring VCO, Delay cell, Current starved ring VCO, Delay, Phase noise.

## I. INTRODUCTION

Oscillators are important building blocks of analog and digital systems. A ring oscillator (ROSC) is commonly used for generating sinusoidal signals for various applications such as true random number generator [1], current to voltage converter for temperature sensor [2], aging sensor [3], analog and digital converters [4], [5], phase-locked loop [6], for calibrating baseline interconnect path [7]. A ROSC is preferred because it has a robust and compact design and a wide tuning range. It consists of several necessary delay cells (d-cells). The performance of d-cells directly affects the quality of ROSC's output. In general, to get a good quality sinusoidal clock signal, a d-cell can be improved. In some applications, differential signals are preferred over single-ended to reduce power supply noise [4], [8]. In those applications, the d-cells of the ROSC can also be differential in structure [2], [4], [5]. Based on DDC, there are two types of ROSCs. 1) dual ROSCs using CMOS inverters as the main block and connected back-to-back auxiliary CMOS inverters at each node of main CMOS inverters connecting both the ROSCs to get complementary differential outputs [4], [5]. 2), latch-based DDC [8], [9]. Both of the ROSCs topologies are equally used in

different applications [4], [5], [8]. In this paper, a differential CSRVCO is presented.

Free-running ROSCs are used in some applications [1], [3]. In most of the applications, tuning of frequency is required [2], [4]–[10]. The tuning of frequency can be done by connecting different capacitive loads at the output node [6] or by varying supply current [2], [4], [5], [7]–[9]. The large area of capacitors is one of the drawbacks of the capacitive load-based frequency tuning technique. Moreover, to select different capacitors for frequency tuning, a controlled switch logic is required [6], which adds to additional complexity in circuitry. On the other hand, current-controlled ROSC can be designed such that by varying a control voltage, the supply current can be adjusted for frequency tuning [2]. In general, current control ROSC using control voltage for frequency tuning is simpler to implement in layout [11].

In this paper, a DDC is presented and implemented in a CSRVCO. The CSRVCO consumes low power and occupies less area. Additionally, it provides less jittery sinusoidal outputs while maintaining all other specifications compared to the literature's available designs. This is the first time a pass transistor gate-based d-cell in a ROSC is presented to the best of the author's knowledge. The paper is organized in the following manner: Section II describes the CSRVCO using the proposed DDC. Section III shows a mathematical analysis of propagation delays in the DDC. Section IV shows post-layout simulation results for the CSRVCOs using the proposed DDCs, and Section V concludes the paper.

## II. THE RING VOLTAGE CONTROLLED OSCILLATOR USING THE PROPOSED DIFFERENTIAL DELAY CELL

Delay cells (d-cells) are an essential part of a ROSC. The number of d-cells decides the total number of stages of a ROSC and is connected based on clock phase requirement. In this paper, a 3-stage CSRVCO using three DDCs is presented to verify the operation of the proposed DDCs as shown in Fig. 1.

The CSRVCO uses three proposed DDCs connected with a current mirror (MOSFET  $M_9$ - $M_{10}$ ) to control the current using a control voltage ( $V_{ctrl}$ ) using MOSFET  $M_7$ . A diode connected to MOSFET  $M_8$  is connected at the source of the MOSFET  $M_7$  to increase the linearity. In the proposed DDC,

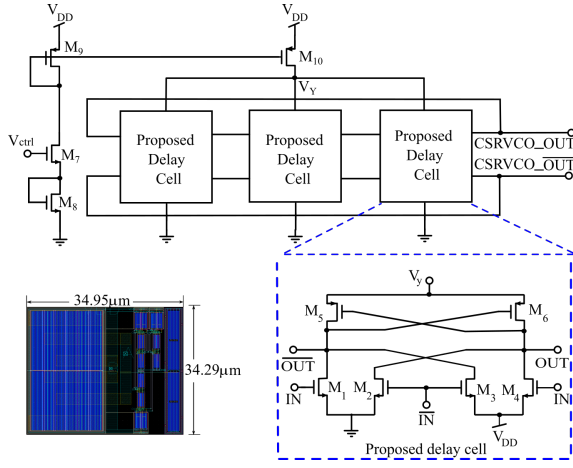


Fig. 1. The 3-stage CSRVCO using the proposed DDCs occupying area of  $1198.43\mu\text{m}^2$ .

transistors ( $M_1, M_2, M_3, M_4$ ) form a differential switch. In the differential switch, the source terminal of transistors ( $M_1-M_2$ ) is connected to the ground, and the source terminal of transistors ( $M_3-M_4$ ) is connected to supply voltage ( $V_{DD}$ ). Each of the transistors ( $M_1, M_2, M_3, M_4$ ) is connected to form a pass transistor logic. In the CSRVCO (Fig. 1), source of  $M_5$  and  $M_6$  are connected to the drain of MOSFET  $M_{10}$  that is at  $V_Y$  voltage level. The total area occupied by CSRVCO is  $1198.43\mu\text{m}^2$ .

A switching operation in the DDCs is shown in Fig. 2. Generally, the input can be switched from voltage logic 'high' to 'low' or 'low' to 'high'. In accordance with the switching voltage levels, both cases are explained below:

A. When input 'IN' switches to logic 'high' and input 'IN-bar' switches to logic 'low' at  $t = t_1$ :

At time  $t = t_1$ , 'OUT' is at logic 'high' level ( $V_Y \approx V_{DD}$ ) and 'OUT-bar' at logic 'low' level (GND). When input 'IN' becomes high at  $t = t_1$ , MOSFETs ( $M_1$  &  $M_4$ ) start conducting, and MOSFETs ( $M_2$  &  $M_3$ ) are in 'off' state. As MOSFET  $M_4$  turns on, the capacitor  $C_1$  starts to charge as shown in Fig. 2(a). The capacitor charges up to the latch threshold voltage level required to initiate positive feedback action. Then, PMOS based positive feedback circuit (MOSFET  $M_5$  &  $M_6$ ) pulls the voltage at OUT to the source potential of the PMOS transistors. On the other side, MOSFET  $M_1$  starts to conduct and at  $t = t_1$ , capacitor  $C_2$  discharges. Both the capacitors ( $C_1$  &  $C_2$ ) are load capacitance that are equal in general. In this paper, capacitors ( $C_1$  &  $C_2$ ) are mentioned differently only for concept explanation.

B. When input 'IN-bar' switches to logic 'high' and input 'IN' switches to logic 'low' at  $t = t_1 + \frac{T}{2}$ :

In time  $\frac{T}{2}$ , output 'OUT' charges to  $V_Y$  and output OUT discharges to zero (ground potential). At  $t = t_1 + \frac{T}{2}$ , input 'IN' is at 'high' logic level ( $V_{DD}$ ) and input IN is at 'low' logic level (ground potential). MOSFET ( $M_2$  &  $M_3$ ) start conducting, and MOSFET ( $M_1$  &  $M_4$ ) are at cutoff state.

As MOSFET  $M_3$  turns on, the capacitor  $C_2$  starts to charge as shown in Fig. 2(b). As output voltage reaches the latch threshold voltage of positive feedback, it acts and pulls the voltage of node 'OUT' to  $V_Y$ . Similarly, when MOSFET  $M_2$  begins to conduct, then capacitor  $C_1$  discharges through MOSFET  $M_2$ . A detailed mathematical analysis of delay time calculation is given in the next section.

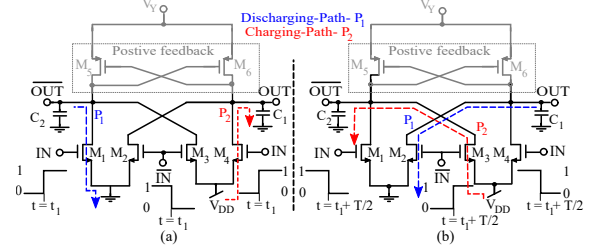


Fig. 2. The proposed DDC depicts the charging and discharging paths from a capacitive load, ensuring a single NMOS transistor in each path to get better jitter performance at the output for different switching combinations at the input.

In the proposed DDC, a single NMOS transistor comes in a charging and discharging path from output capacitance during input-output switching activities. NMOS transistors are bad switches for transmitting 'high' logic. However, the drawback would not affect the operation of the proposed DDC because a minimum latch threshold voltage is required to activate positive feedback during latch operation that switches the voltage levels to a maximum or minimum voltage, depending on voltage logic. In the proposed DDC, PMOS transistors ( $M_5$  &  $M_6$ ) are connected in a back-to-back fashion, providing positive feedback. Thus, the voltage level at the drain terminal of PMOS transistors latches the pass transistor's output to a maximum or minimum voltages based on output voltage logic [12].

TABLE I  
COMPARISON BETWEEN PSEUDO-DIFFERENTIAL AND PROPOSED DDC BASED RING VCOS.

| Parameters                                 | Pseudo-Differential Ring VCO [13] | Proposed DDC based Ring VCO |
|--------------------------------------------|-----------------------------------|-----------------------------|
| Techn. (nm)                                | 65                                | 65                          |
| Transistors per Delay Cell                 | 8                                 | 6                           |
| Frequency (GHz)                            | 2.76                              | 2.76                        |
| Supply Voltage (V)                         | 1                                 | 1                           |
| PN@1MHz ( $\frac{\text{dBc}}{\text{Hz}}$ ) | -92                               | -95.12                      |
| Power (mW)                                 | 8.47                              | 2.62                        |
| FoM  ( $\frac{\text{dBc}}{\text{Hz}}$ )    | 151                               | 159                         |

Generally, pseudo-differential ring oscillator [13] is used for differential VCO design in CMOS logic. The proposed DCC-based VCO is compared with the pseudo-differential ring oscillator operating at a similar operating frequency. It is observed that the proposed DCC-based VCO has better phase noise, power, figure of merit (FOM), and lesser number of transistors shown in Table I. The propagation delay of each stage can be controlled by varying currents which is explained in the next section.

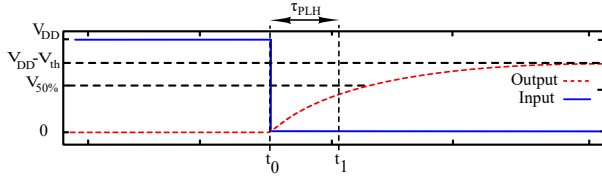


Fig. 3. Input and output responses showing  $\tau_{PLH}$  for the CSRVC

### III. ANALYSIS FOR PROPAGATION DELAY IN THE PROPOSED DELAY CELL

In digital switching circuits, the propagation delay  $\tau_P$  is calculated as:

$$\tau_P = \frac{\tau_{PHL} + \tau_{PLH}}{2} \quad (1)$$

Where ( $\tau_{PHL}$ ) is high to low delay time and ( $\tau_{PLH}$ ) is low to high delay time. The maximum frequency can also be calculated by taking the inverse of the propagation delay time. A detailed analysis for ( $\tau_{PHL}$ ) and ( $\tau_{PLH}$ ) are given below:

#### A. Calculation of low to high propagation delay time ( $\tau_{PLH}$ )

Let's assume input IN is at a 'high' logic level and moves to a 'low' logic level, and the output OUT is at a 'low' logic level initially, as shown in Fig. 3. As per the proposed DDC operation mentioned in Section II for the charging path using MOSFET  $M_4$ , the output OUT starts moving from logic 'low' to 'high'. In the process, output load capacitance starts charging using turned-on MOSFET  $M_4$  as shown in Fig. 2(a). Since MOSFET  $M_4$  is in saturation region by the given condition, it starts to charge load capacitance with the saturation current  $I_{D,sat}$ . The short-channel effects are ignored to simplify the analysis.

$$I_{D,sat} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = C_1 \frac{dV_{C,sat}}{dt} \quad (2)$$

where,  $\mu_n$  = Mobility of electron;  $C_{ox}$  = Oxide capacitance; W and L = width and length of MOSFET;  $V_{GS}$  = Gate to source voltage;  $V_{th}$  = Threshold voltage;  $C_1$  = Load capacitance;  $V_{C,sat}$  = Maximum voltage capacitance.

Now, perform the integration after simplifying the equation (2) and assume that output is at voltage level  $V_{OUT}$ .

$$\int_0^t dt = \frac{2C_1}{\mu_n C_{ox} \frac{W}{L}} \int_0^{V_{OUT}} \frac{dV_{OUT}}{(V_{DD} - V_{OUT} - V_{th})^2} \quad (3)$$

Integrate equation 3 and it gives:

$$\tau_{PLH} = \frac{2C_1}{\mu_n C_{ox} \frac{W}{L}} \left( \frac{1}{(V_{DD} - V_{OUT} - V_{th})} - \frac{1}{V_{DD} - V_{th}} \right) \quad (4)$$

Equation (4) can provide charging characteristic to the output capacitance and finally settles at final value of  $V_{DD} - V_{th}$ . However, positive feedback activates at time instant before reaching the final values.

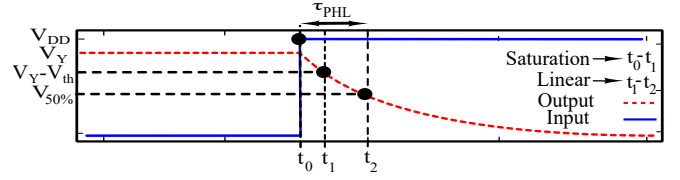


Fig. 4. Input and output responses showing  $\tau_{PHL}$  for the CSRVC

#### B. Calculation of high to low propagation delay time ( $\tau_{PHL}$ )

Discharging path of the load capacitance is investigated to find  $\tau_{PHL}$  as shown in Fig. 2(b). Let's assume the capacitor is fully charged with voltage level  $V_Y$ , that is, the source terminal's voltage of PMOS transistors in Fig. 1. After the regenerative action of positive feedback,  $V_Y$  is approximately equal to  $V_{DD}$ . Initially, MOSFET  $M_1$  (Fig. 2(a)) is in the saturation region because the voltage ( $V_Y \approx V_{DD}$ ) at the drain to the source terminal is equal to the voltage ( $V_Y$ ) at the gate to the source terminal. Now output starts to move from logic 'high' to 'low' as shown in Fig. 4. The charge on the capacitor starts to decrease, leading to a voltage drop reduction. When the voltage drops less than ( $V_Y - V_{th}$ ), then the MOSFET changes the operating region from saturation to linear region. The detailed analysis based on each of the operating regions is mentioned below.

1) *The time period  $t_0$  to  $t_1$  (Saturation region):* The time  $t_0$  to  $t_1$ , where MOSFET  $M_1$  (Fig. 2(a)) is in the saturation region is given by

$$\int_{t_0}^{t_1} dt = - \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_Y - V_{th})^2} \int_{V_Y}^{V_Y - V_{th}} dV_{OUT} \quad (5)$$

The time difference between  $t_0$  to  $t_1$  after integration is given by

$$t_1 - t_0 = \frac{2C_2 V_{th}}{\mu_n C_{ox} \frac{W}{L} (V_Y - V_{th})^2} \quad (6)$$

The description of the parameters is the same as previous definitions mentioned during  $\tau_{PLH}$  calculations.

2) *After time instant  $t = t_1$  (Linear region):* After time instant  $t = t_1$ , MOSFET  $M_1$  (Fig. 2(a)) enters into linear region. The  $t_2$  is time instant at which voltage reach 50% of maximum output voltage ( $V_{OH} = V_Y - V_{th}$ ). The time  $\tau_{PHL}$  at which the output voltage discharges upto 50% of  $V_{OH}$  i.e.  $V_{50\%}$ ) is calculated below:

$$\int_{t=t_1}^{t=t_2} dt = -2C_2 \int_{V_{OH}}^{V_{50\%}} \frac{1}{\mu_n C_{ox} \frac{W}{L} (2V_{OH} V_{OUT} - V_{OUT}^2)} dV_{OUT} \quad (7)$$

$$t_2 - t_1 = \frac{C_2}{\mu_n C_{ox} \frac{W}{L} (V_{OH})} \ln \left( \frac{2(V_{OH}) - V_{50\%}}{V_{50\%}} \right) \quad (8)$$

$\tau_{PHL}$  can be calculated by adding equation 6 and 8:

$$\tau_{PHL} = \frac{C_2}{\mu_n C_{ox} \frac{W}{L} (V_{OH})} \left[ \frac{2V_{th}}{(V_{OH})} + \ln \left( \frac{4(V_{OH})}{V_Y} - 1 \right) \right] \quad (9)$$

In the next section, equations 4 and 9 are plotted and compared with simulation results. A positive feedback, designed using PMOS transistors ( $M_5$  and  $M_6$ ), is connected at the outputs

( $\overline{OUT}$  and  $\overline{OUT}$ ) as shown in Fig 1. The feedback ensures voltage levels ' $V_Y \approx V_{DD}$ ' and 'GND' at the outputs [12]. The next section shows post-layout simulation results for the proposed DDC-based CSRVC.

#### IV. POST LAYOUT SIMULATION RESULTS

The proposed DDC is designed in a 65 nm CMOS technology. Figure 5 shows output response of the proposed DDC by plotting mathematical equations (4) and (9) for  $\tau_{PHL}$  &  $\tau_{PLH}$  and comparing with simulation results without adding effect of positive feedback. These simulations are performed in Typical-Typical (TT) corner at 27°C. The required values are used from circuit and extracted by simulations for different parameters in equation (4) and (9). Simulation results for  $\tau_{PHL}$  and  $\tau_{PLH}$  match with equation (4) and (9) plots.

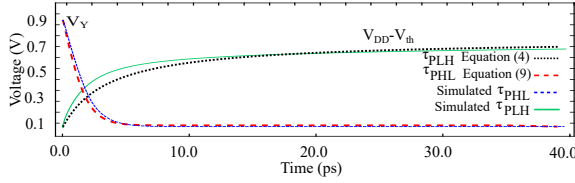


Fig. 5. Simulation results of output voltage of the proposed DDC in time domain ( $\tau_{PHL}$  and  $\tau_{PLH}$ ) are similar to plot using equations 4 and 9.

The CSRVC using the proposed DDC has been laid out in 65nm CMOS technology as shown in Fig. 1. Post-layout simulation results validating the functionality of the CSRVC are shown in Fig. 7. The total peak-to-peak jitter is 17.01ps at a frequency of 2.76GHz. Jitter can further be improved by adding additional circuitry that changes the threshold level [14], [15]. Phase noise (Fig. 7) is  $-94.54 \frac{dBc}{Hz}$

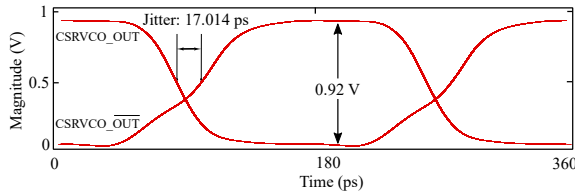


Fig. 6. Eye diagram shows 2.76GHz sinusoidal signal with eye-opening of 0.92mV and peak to peak jitter of 17.01 ps.

at 1 MHz of offset frequency. Output frequency range (control voltage with frequency) is from 0.88 GHz to 2.76 GHz and  $K_{VCO}$  is  $6.55 \frac{GHz}{V}$  as shown in Fig 8. In addition, Fig. 9 shows the CSRVC works fine with variation of process parameter, temperature and supply voltage for control voltage of 1 V.

A comparison between proposed and published ROSCs is shown in Table II. The phase noise performance of the proposed ring VCO is better than the architectures given in references [16], [18], [20]. The phase noise is comparable with references [17], [19]. The power dissipation for the architectures given in references [16] is lower because of the lower technology node. In References [17], and [18], ROSCs using inverter d-cells are designed for eight and five stages,

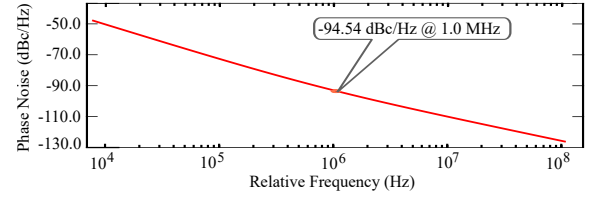


Fig. 7. Phase noise is  $-94.54 \frac{dBc}{Hz}$  for the CSRVC at offset of 1 MHz.

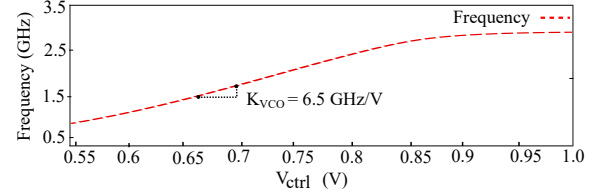


Fig. 8. Variation in frequency and  $K_{VCO}$  with control voltage shows linear tuning range from 1.16 GHz to 2.25 GHz.

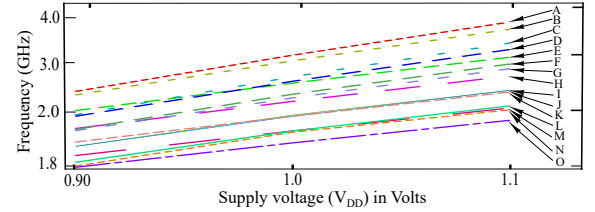


Fig. 9. Change in frequency with PVT variations keeping control voltage of 1 V (A = {FF, 0°C}, B = {FF, 27°C}, C = {FNTP, 0°C}, D = {FNTP, 27°C}, E = {FF, 125°C}, F = {TT, 0°C}, G = {TT, 27°C}, H = {FNTP, 125°C}, I = {SNTP, 0°C}, J = {SNTP, 27°C}, K = {TT, 125°C}, L = {SS, 27°C}, M = {SNTP, 125°C}, N = {SS, 0°C}, and O = {SS, 125°C}).

TABLE II  
COMPARISON BETWEEN PROPOSED AND OTHER PUBLISHED ROSCs

|                              | [16]        | [17]       | [18]   | [19]     | [20]       | This Work  |
|------------------------------|-------------|------------|--------|----------|------------|------------|
| Delay Stages                 | 4           | 8          | 5      | 13       | 4          | 3          |
| Techn. (nm)                  | 16          | 65         | 65     | 28       | 180        | 65         |
| Supply (V)                   | 0.5         | 1          | 1.2    | 1        | 1.8        | 1          |
| Max. Freq. (GHz)             | 2.53        | 3.11       | 2      | 1.76     | 2.53       | 2.76       |
| Frequency Range (GHz)        | 0.056 -0.33 | 6.68 -3.56 | 1.2 -2 | 0.7 -2.7 | 1.78 -2.53 | 0.88 -2.76 |
| PN@1MHz ( $\frac{dBc}{Hz}$ ) | -69.2       | -95.1      | -79.1  | -95.7    | -92.6      | -94.5      |
| Area ( $mm^2$ )              | -           | 0.0017     | 0.0007 | 0.008    | 0.0149     | 0.0011     |
| Power (mW)                   | 0.003       | 3.23       | 1.1    | 1.1      | 28         | 5.19       |
| [FoM] ( $\frac{dBc}{Hz}$ )   | 145#        | 159*       | 144*   | 160*     | 146*       | 156#       |

\* Measured, # Simulated. FoM - The higher the better.

$$FoM = PN - 20 \log \left( \frac{f_{osc}}{f_{offset}} \right) + 10 \log \left( \frac{P_{diss}}{1mW} \right) [19]$$

and in References [19], a pseudo-differential ring VCO with a double tuning approach is used in 28nm FD-SOI CMOS technology. In References [20], CML ring VCO with four stages is used in 180nm CMOS technology. In the proposed DDC-based CSRVC, area occupation and power dissipation are comparatively less than ROSC of references [20] and comparable with ROSC of references [17]. FoM is still comparable

to ROSCs of references [16]–[20].

## V. CONCLUSION

A CSRVCO using pass transistor-based delay cells has been demonstrated. The proposed delay cell has simple circuitry and is easy to design. The architecture is symmetric, which helps in getting a less jittery output response. Post-layout simulation results of the 3-stage CSRVCO confirmed the proper operation of the proposed delay cells. The proposed CSRVCO works from 884 MHz to 2.76 GHz frequency with 1 V supply. The power dissipation of the proposed DDC-based CSRVCO is 5.19 mW for 1 V control voltage at a center frequency of 2.76 GHz. The proposed CSRVCO is very useful in Bluetooth front-end applications.

## REFERENCES

- [1] Keke Gai, Yaoling Ding, An Wang, Liehuang Zhu, Kim-Kwang Raymond Choo, Qi Zhang and Zhuping Wang, "Attacking the Edge-of-Things: A Physical Attack Perspective" in *IEEE Internet of Things Journal (early access)*, 2021, doi: 10.1109/JIOT.2021.3109917.
- [2] B. Zambrano, E. Garzón, S. Strangio, F. Crupi and M. Lanuzza, "A 0.05 mm<sup>2</sup>, 350 mV, 14 nW Fully-Integrated Temperature Sensor in 180-nm CMOS" in *IEEE Transactions on Circuits and Systems II: Express Briefs (early access)*, 2021, doi: 10.1109/TCSII.2021.3112812.
- [3] T. Alnuayri, S. Khurshed, A. L. H. Martínez and D. Rossi, "Differential Aging Sensor Using Subthreshold Leakage Current to Detect Recycled ICs" in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems (early access)*, 2021, doi: 10.1109/TVLSI.2021.3115247.
- [4] A. Babaie-Fishani and P. Rombouts: "Highly linear VCO for use in VCO-ADCs," *Electronics Letters*, 2016, 52, (4), p. 268-270, DOI: 10.1049/el.2015.3807.
- [5] K. M. Al-Tamimi, K. El-Sankary and Y. Fouzar, "VCO-Based ADC With Built-In Supply Noise Immunity Using Injection-Locked Ring Oscillators" in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 66, no. 7, pp. 1089-1093, July 2019, doi: 10.1109/TCSII.2018.2875867.
- [6] Shiheng Yang, Jun Yin, Tailong Xu, Taimo Yi, Pui-In Mak, Qiang Li, and Rui P. Martins, "A 600-μm<sup>2</sup> Ring-VCO-Based Hybrid PLL Using a 30-μW Charge-Sharing Integrator in 28-nm CMOS," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 9, pp. 3108-3112, Sept. 2021, doi: 10.1109/TCSII.2021.3096193.
- [7] R. Saligram, S. Datta and A. Raychowdhury, "Scaled Back End of Line Interconnects at Cryogenic Temperatures," in *IEEE Electron Device Letters*, vol. 42, no. 11, pp. 1674-1677, Nov. 2021, doi: 10.1109/LED.2021.3117277.
- [8] J. Hamilton, S. Yan and T. R. Viswanathan, "A Discrete-Time Input ΔΣ ADC Architecture Using a Dual-VCO-Based Integrator" in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 57, no. 11, pp. 848-852, Nov. 2010, doi: 10.1109/TCSII.2010.2068111.
- [9] I. Kovacs and M. Neag, "New dual-loop topology for ring VCOs based on latched delay cells," *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2018, pp. 1-5, doi: 10.1109/ISCAS.2018.8351757.
- [10] X. Ji, Y. Wang, X. Xia and Y. Guo, "A Capacitively Coupled Noise Circulating VCO" in *IEEE Microwave and Wireless Components Letters*, vol. 31, no. 10, pp. 1127-1129, Oct. 2021, doi: 10.1109/LMWC.2021.3101287.
- [11] M. Schramme, L. Van Brandt, D. Flandre and D. Bol, "Comprehensive Analytical Comparison of Ring Oscillators in FDSOI Technology: Current Starving Versus Back-Bias Control," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 69, no. 5, pp. 1883-1895, May 2022, doi: 10.1109/TCSI.2022.3144527.
- [12] B. Razavi, "The Cross-Coupled Pair - Part II [A Circuit for All Seasons]" in *IEEE Solid-State Circuits Magazine*, vol. 6, no. 4, pp. 9-12, Fall 2014, doi: 10.1109/MSSC.2014.2352532.
- [13] Y. Frans et al., "A 0.5–16.3 Gb/s Fully Adaptive Flexible-Reach Transceiver for FPGA in 20 nm CMOS," in *IEEE Journal of Solid-State Circuits*, vol. 50, no. 8, pp. 1932-1944, Aug. 2015, doi: 10.1109/JSSC.2015.2413849.
- [14] T. Liang, Z. Fu, H. Liu, K. Liu and Y. Xiao, "Minimizing the Jitter of Duty Cycle Distortion Correction Technology Based on Cross Point Eye Diagram Correction," in *IEEE Access*, vol. 7, pp. 106238-106247, 2019, doi: 10.1109/ACCESS.2019.2931474.
- [15] Joonsuk Lee and Beomsup Kim, "A low-noise fast-lock phase-locked loop with adaptive bandwidth control" in *IEEE Journal of Solid-State Circuits*, vol. 35, no. 8, pp. 1137-1145, Aug. 2000, doi: 10.1109/4.859502.
- [16] S. M. I. Huq, O. L. Baroi, S. A. Shihab and S. N. Biswas, "Comparative Study and Design of Current Starved Ring Oscillators in 16 nm Technology" in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 4, pp. 1098-1102, April 2021, doi: 10.1109/TCSII.2020.3027897.
- [17] I. F. Sun, J. Yin, P. I. Mak and R. P. Martins, "A Comparative Study of 8-Phase Feedforward-Coupling Ring VCOs" in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 66, no. 4, pp. 527-531, April 2019, doi: 10.1109/TCSII.2018.2865465.
- [18] M. Kim, S. Choi, T. Seong and J. Choi, "A Low-Jitter and Fractional-Resolution Injection-Locked Clock Multiplier Using a DLL-Based Real-Time PVT Calibrator With Replica-Delay Cells" in *IEEE Journal of Solid-State Circuits*, vol. 51, no. 2, pp. 401-411, Feb. 2016, doi: 10.1109/JSSC.2015.2496781.
- [19] D. Gaidioz, M. De Matos, A. Cathelin and Y. Deval, "Ring VCO Phase Noise Optimization by Pseudo-Differential Architecture in 28nm FD-SOI CMOS," 2020 *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2020, pp. 1-4, doi: 10.1109/ISCAS45731.2020.9180557.
- [20] X. Gui, R. Tang, Y. Zhang, D. Li and L. Geng, "A Voltage-Controlled Ring Oscillator With VCO-Gain Variation Compensation" in *IEEE Microwave and Wireless Components Letters*, vol. 30, no. 3, pp. 288-291, March 2020, doi: 10.1109/LMWC.2020.2967391.